Industrial IoT & Embedded Systems Engineer Portofolio

- Vehicle Monitoring System (CAN Bus SAE J1939)
- Industrial Energy Minitoring System (Modbus RTU)

Industrial IoT & Embedded Systems Engineer

CAN Bus SAE J1939 • Modbus RTU • MQTT • Python • InfluxDB • Grafana • REST API

End To End IoT Solution For Telematics & Industrial Monitoring

Marwan Saputra

Tentang Saya

Saya memiliki pengalaman mengembangkan sistem monitoring berbasis IoT untuk berbagai kebutuhan, mulai dari telematika kendaraan menggunakan CAN Bus SAE J1939 hingga monitoring energi dan facility berbasis Modbus RTU. Sistem yang saya bangun mencakup komunikasi perangkat, pengolahan data, backend Python, database time-series, dashboard monitoring, dan REST API.

Keahlian Teknis

Industrial Communication

- CAN Bus SAE J1939
- Modbus RTU
- RS485
-

Embedded Systems

- ESP32
- Arduino IDE
- Embedded Programming

IoT & Networking

- MQTT TLS
- EMQX Cloud

Backend Development

- Python & Flask
- REST API
- ngrok

Database & Visualization

- InfluxDB Cloud
- Grafana

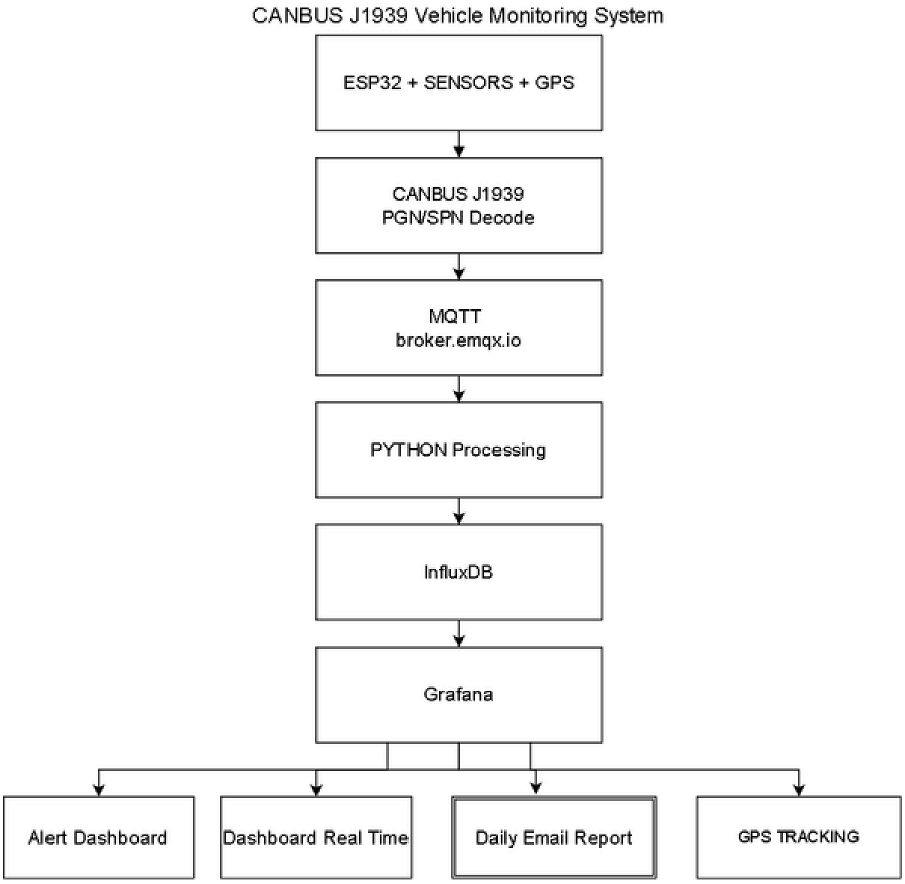
Hardware & Sensors

- MCP2515 CAN Controller . PZEM-004T . DHT22 . MQ-2 Gas Sensor . GPS NEO-6M

PROJECT 1

VEHICLE MONITORING SYSTEM
CAN BUS SAE J1939
Overview

IoT Vehicle Monitoring System merupakan Vehicle Monitoring System berbasis ESP32 dan CAN Bus SAE J1939 yang mendukung implementasi Fleet Management melalui pemantauan data kendaraan secara real-time, mengirim data melalui MQTT, menyimpan data ke InfluxDB, menampilkan dashboard Grafana, serta menyediakan Telegram Alert dan Daily Email Report.



Technologies Used

- **ESP32 + MCP2515 + GPS NEO-6M + Potensiometer**
- **CAN Bus SAE J1939**
- **MQTT (EMQX Broker)**
- **Python**
- **InfluxDB**
- **Grafana**
- **Dashboard Real-Time**
- **Alert Dashboard**
- **Daily Email Report**

CAN Bus SAE J1939 Decoding

Sistem menggunakan ESP32 yang terhubung dengan MCP2515 CAN Bus Controller untuk menerima frame data CAN Bus SAE J1939 dari kendaraan. Data CAN kemudian diproses melalui PGN/SPN Decoding sehingga menghasilkan parameter seperti Engine RPM, Engine Coolant Temperature, Fuel Level, Battery Voltage, dan data telemetri lainnya sebelum dikirim melalui MQTT ke sistem monitoring.

Technical Implementation

- ESP32 + MCP2515 CAN Bus Interface + GPS NEO 6-M
- CAN Bus SAE J1939 Communication
- PGN/SPN Decoding
- Engine RPM Monitoring
- Engine Coolant Temperature
- Fuel Level Monitoring
- Latitude, Longitude, Kecepatan, Satelit
- MQTT Data Publishing
- Python Data Processing

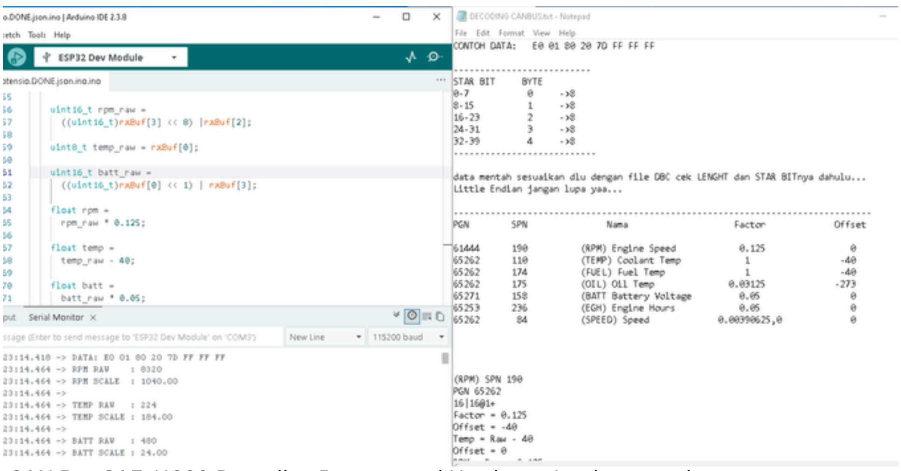
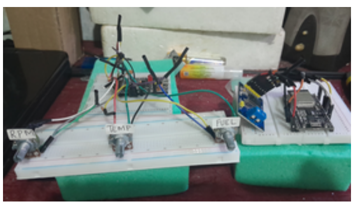
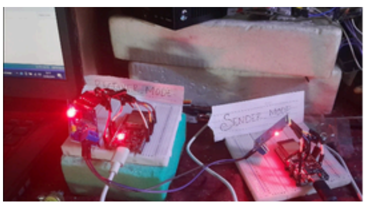


Figure 2. CAN Bus SAE J1939 Decoding Process and Hardware Implementation.

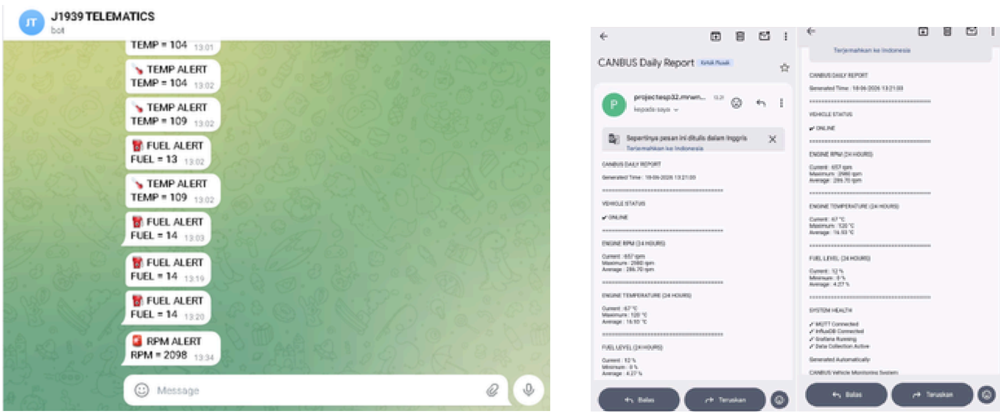

ESP32 + Potensiometer Sensor Simulation


ESP32 + MCP2515 CAN Bus Module


ESP32 + GPS NEO-6M Module

Monitoring Results & Notification System

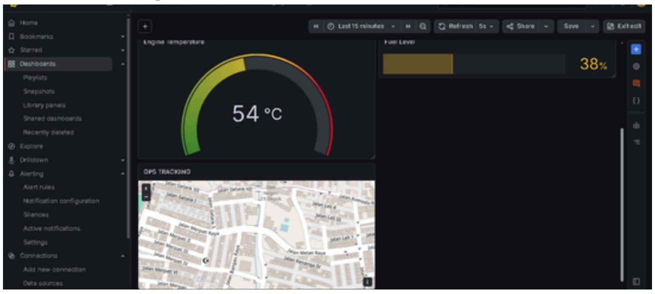
Figure 3. Telegram Alert & Daily Email Report.



Sistem menyediakan notifikasi Telegram secara otomatis ketika parameter kendaraan melebihi batas yang ditentukan. Selain itu, sistem menghasilkan Daily Email Report yang berisi ringkasan data kendaraan untuk mendukung aktivitas monitoring dan evaluasi operasional.

Figure 4.
Real-Time Monitoring Dashboard

Gps Tracking Dashboard



- Engine RPM Monitoring
- GPS Tracking
- Engine Coolant Temperature
- Real-Time Visualization
- Fuel Level Monitoring
- Historical Data Monitoring
- Latitude, Longitude, Kecepatan, Satelit
- Alert Dashboard

Kesimpulan

Project ini berhasil mengimplementasikan sistem monitoring kendaraan end-to-end mulai dari akuisisi data CAN Bus, pemrosesan data, hingga visualisasi dashboard serta notifikasi secara real-time.

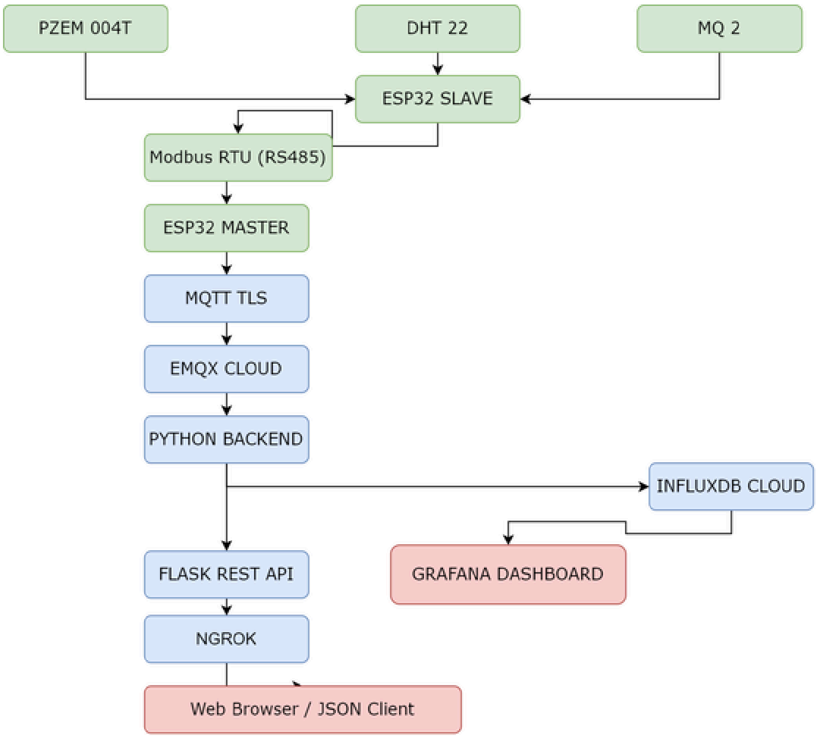
PROJECT 2

Industrial Energy Monitoring System

Modbus RTU • RS485 • MQTT TLS

Overview

Industrial Energy Monitoring System merupakan sistem monitoring energi dan kondisi lingkungan berbasis Internet of Things (IoT) yang menggunakan komunikasi Modbus RTU melalui RS485 untuk mengumpulkan data dari berbagai sensor. Data kemudian dikirim secara aman menggunakan MQTT TLS ke EMQX Cloud, diproses menggunakan Python, disimpan pada InfluxDB Cloud, divisualisasikan melalui Grafana Dashboard, serta menyediakan REST API untuk kebutuhan integrasi aplikasi.



Technologies Used

- ESP32 Master & ESP32 Slave
- PZEM-004T Energy Meter
- DHT22 Temperature & Humidity
- Sensor MQ-2 Gas
- Modbus RTU (RS485)
- MQTT TLS (EMQX Cloud)
- Python
- InfluxDB Cloud
- Grafana
- Flask REST API
- ngrok
- Real-Time Dashboard
- REST API Integration

Modbus RTU Communication

Sistem menggunakan ESP32 Master yang berkomunikasi dengan ESP32 Slave melalui Modbus RTU (RS485) untuk memperoleh data dari sensor PZEM-004T, DHT22, dan MQ-2. Data yang diterima kemudian dipublikasikan menggunakan MQTT TLS ke EMQX Cloud, diproses oleh aplikasi Python, sebelum disimpan pada InfluxDB Cloud untuk divisualisasikan melalui Grafana Dashboard dan diakses melalui REST API.

Technical Implementation

- ESP32 Master & ESP32 Slave
- Modbus RTU (RS485) Communication
- PZEM-004T Energy Monitoring
- DHT22 Temperature & Humidity Monitoring
- MQ-2 Gas Monitoring
- MQTT TLS Data Publishing (EMQX Cloud)
- Python Data Processing
- InfluxDB Cloud Storage
- Grafana Dashboard
- REST API

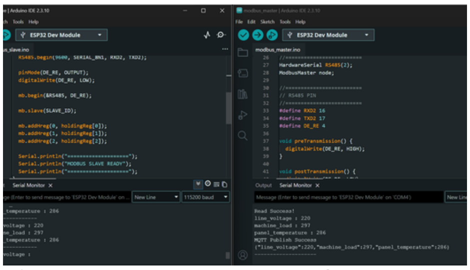


Figure 5. ESP32 Master–Slave Modbus RTU Communication Implementation

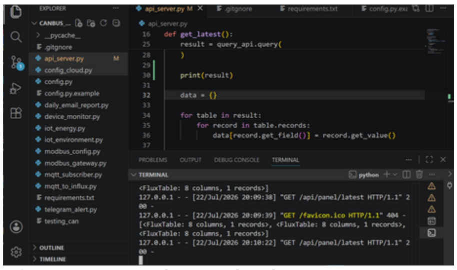


Figure 6. Python Flask REST API Implementation and Data Retrieval from InfluxDB Cloud.

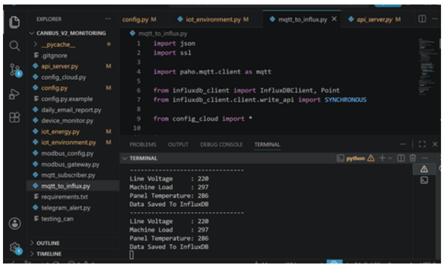


Figure 7. MQTT TLS Data Processing and InfluxDB Cloud Storage.

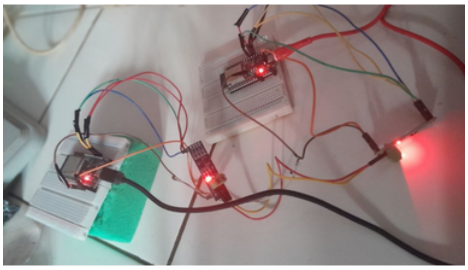


Figure 8. ESP32 Master, ESP32 Slave, and RS485 Modbus RTU Hardware Implementation.

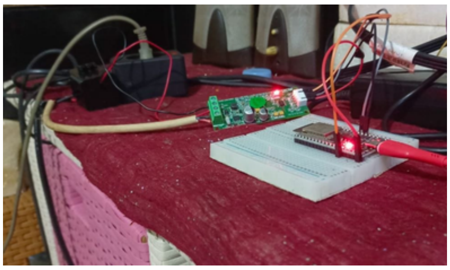


Figure 9. PZEM 004T Sensor Hardware Implementation.

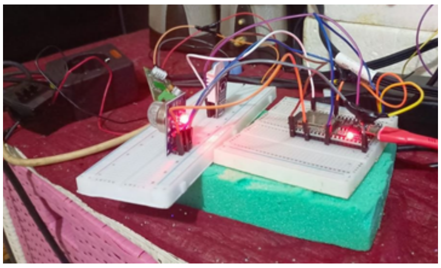


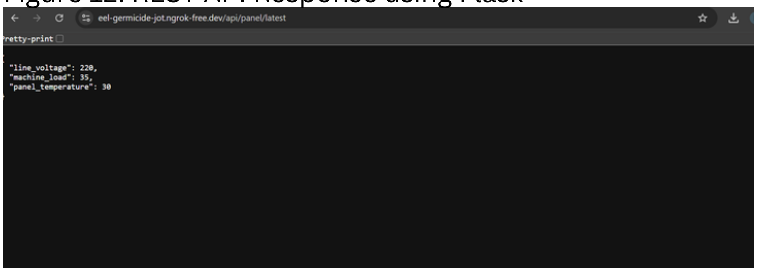
Figure 10. DHT 22 and MQ-2 Sensor HardwareImplementation.

Cloud Dashboard & REST API

Figure 11. Grafana cloud Dashboard.

- Dashboard Energy
- Voltage • Current • Power • Energy
- Temperature
- Gas

Figure 12. REST API Response using Flask



Sistem menyediakan REST API berbasis Flask yang memungkinkan data monitoring diakses dalam format JSON untuk integrasi dengan aplikasi web, mobile, maupun sistem pihak ketiga.

Project Summary

Project Achievement

- ESP32 Master & Slave Communication
- Modbus RTU over RS485
- PZEM-004T Energy Monitoring
- DHT22 & MQ-2 Monitoring
- MQTT TLS (EMQX Cloud)
- Python Backend Processing
- InfluxDB Cloud Storage
- Grafana Dashboard
- Flask REST API
- ngrok Public Endpoint

Conclusion

This project successfully implements an end-to-end Industrial Energy Monitoring System using ESP32 Master–Slave communication, Modbus RTU over RS485, MQTT TLS, Python backend, InfluxDB Cloud, Grafana Dashboard, and Flask REST API. The system enables real-time monitoring of electrical parameters and environmental conditions while providing secure cloud connectivity and data integration for Industrial IoT applications.

This project demonstrates practical experience in Embedded Systems, Industrial Communication, Cloud IoT, Backend Development, and Data Visualization for Industrial IoT applications.

github profile

Open Github

github project V1

View Repository

github project V2

View Repository

Scan the QR Codes or click the links below to access the complete source code and project documentation.